

PHYSICS EXAM NOTES

Serway & Jewett — Physics for Scientists & Engineers, 9th Ed.

Units 1–4 | Kinematics · Dynamics · Electrostatics · Circuits

Unit 1	Kinematics & Vectors (Ch 1–4)	Unit 3	Electrostatics (Ch 23–25)
Unit 2	Dynamics & Energy (Ch 5–8)	Unit 4	Capacitors & DC Circuits (Ch 26–28)

UNIT 1 — KINEMATICS & VECTORS

Chapters 1 · 2 · 3 · 4

Chapter 1 — Physics & Measurement

Physics rests on carefully defined base quantities. Everything else is derived from them.

- **SI base units:** metre (m), kilogram (kg), second (s), ampere (A), kelvin (K), mole (mol), candela (cd).
- **Dimensional analysis:** check that both sides of an equation carry the same dimensions — a powerful error-catching tool. Dimensions: [L] length, [M] mass, [T] time.
- **Significant figures:** result has as many sig-figs as the least precise input. Exact numbers (counting) have infinite sig-figs.
- **Order-of-magnitude estimation:** round all quantities to the nearest power of 10; useful for reality checks.

Unit conversion chain

Multiply by a conversion factor = 1 (e.g. 1 m / 100 cm = 1)

Chapter 2 — Motion in One Dimension

One-dimensional kinematics describes straight-line motion using position x , velocity v , and acceleration a as scalar functions of time.

$x(t)$	Position; displacement $\Delta x = x_f - x_i$
$v_{\text{avg}} = \Delta x / \Delta t$	Average velocity (vector; can be negative)
$v = dx/dt$	Instantaneous velocity — slope of x - t graph
$a = dv/dt$	Instantaneous acceleration — slope of v - t graph

Kinematic Equations (constant acceleration only)

$$v = v_{\text{initial}} + at$$

$$x = x_{\text{initial}} + v_{\text{initial}}t + \frac{1}{2}at^2$$

$$v^2 = v_{\text{initial}}^2 + 2a(x - x_{\text{initial}})$$

$$x = x_{\text{initial}} + \frac{1}{2}(v_{\text{initial}} + v)t$$

- **Freely falling objects:** $a = -g = -9.80 \text{ m/s}^2$ (downward). The kinematic equations apply with this substitution.
 - **v–t graph:** area under curve = displacement; slope = acceleration.
- ★ These 4 equations only hold when acceleration is **CONSTANT** throughout the interval.

Chapter 3 — Vectors

Vectors have both magnitude and direction. They obey special addition rules.

- **Scalar:** magnitude only (speed, mass, temperature).
- **Vector notation:** bold **A** or arrow over letter. Magnitude $|\mathbf{A}| = A \geq 0$.
- **Component form:** $\mathbf{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ where $A_x = A \cos\theta$, $A_y = A \sin\theta$
- **Magnitude:** $A = \sqrt{A_x^2 + A_y^2}$
- **Direction:** $\theta = \arctan(A_y/A_x)$ — check the quadrant!
- **Vector addition:** $\mathbf{R} = \mathbf{A} + \mathbf{B} \rightarrow R_x = A_x + B_x, R_y = A_y + B_y$
- **Dot product:** $\mathbf{A} \cdot \mathbf{B} = AB \cos\theta = A_x B_x + A_y B_y + A_z B_z$ (scalar result)
- **Cross product:** $|\mathbf{A} \times \mathbf{B}| = AB \sin\theta$ (vector, perpendicular to both)

Chapter 4 — Motion in Two Dimensions

Extend 1D kinematics to 2D by treating x and y components independently. Acceleration in one direction does NOT affect the other.

Projectile Motion ($a_x = 0$, $a_y = -g$)

$$x = x_{\text{initial}} + v_{x_{\text{initial}}} t \text{ (horizontal — constant velocity)}$$

$$y = y_{\text{initial}} + v_{y_{\text{initial}}} t - \frac{1}{2}gt^2 \text{ (vertical — free fall)}$$

$$v_{x_{\text{initial}}} = v_{\text{initial}} \cos\theta, v_{y_{\text{initial}}} = v_{\text{initial}} \sin\theta$$

$$\text{Range } R = \frac{v_{\text{initial}}^2 \sin(2\theta)}{g} \text{ (on level ground)}$$

Uniform Circular Motion

Speed v = constant; direction changes

Centripetal acceleration: $a_c = v^2/r$ (directed inward)

Period: $T = 2\pi r/v$

- **Relative velocity:** $\mathbf{v}_{AB} = \mathbf{v}_{AE} - \mathbf{v}_{BE}$ (velocity of A relative to B, via Earth frame)

★ Max range for projectile on flat ground at $\theta = 45^\circ$. At complementary angles (e.g. 30° and 60°) the range is the same.

UNIT 2 — DYNAMICS & ENERGY

Chapters 5 · 6 · 7 · 8

Chapter 5 — The Laws of Motion

Newton's three laws explain how forces cause — or fail to cause — changes in motion.

Newton's 1st	An object remains at rest or moves at constant velocity unless acted on by a net force. (Law of inertia)
Newton's 2nd	$\Sigma F = ma$ — net force equals mass $\times$ acceleration. Both sides are vectors.
Newton's 3rd	For every action force there is an equal and opposite reaction force. $F_{AB} = -F_{BA}$

- **Free-body diagram (FBD):** Draw the object isolated, with all forces acting ON it. Never include forces it exerts on other objects.
 - **Normal force N:** perpendicular contact force from a surface.
 - **Tension T:** force transmitted through a rope or string (massless rope $\rightarrow$ same T throughout).
 - **Weight W:** $W = mg$ (downward gravitational force, $g = 9.80 \text{ m/s}^2$).
 - **Static friction:** $f_s \leq \mu_s N$ (prevents sliding).
 - **Kinetic friction:** $f_k = \mu_k N$ (opposes sliding motion). $\mu_k < \mu_s$.
- ★ Apply Newton's 2nd separately to each axis: $\Sigma F_x = ma_x$, $\Sigma F_y = ma_y$.

Chapter 6 — Circular Motion & Applications of Newton's Laws

For an object in circular motion, a net inward (centripetal) force is required. There is no separate 'centrifugal force' — it is a fictitious force felt in a rotating frame.

Centripetal Force

$$\Sigma F_c = mv^2/r \text{ (net force toward center)}$$

$$a_c = v^2/r = \omega^2 r \text{ (}\omega = \text{angular speed in rad/s)}$$

- **Non-uniform circular motion:** has both centripetal (a_c) and tangential (a_t) components. Total $a = \sqrt{a_c^2 + a_t^2}$
- **Resistive forces (drag):** $R = bv$ (low speed) or $R = \frac{1}{2}D\rho A v^2$ (high speed). At terminal velocity, drag = weight.
- **Apparent weight:** in an accelerating elevator, apparent weight = $m(g \pm a)$. Going up and speeding up $\rightarrow$ feel heavier.

Chapter 7 — Energy of a System

Energy is the capacity to do work. Work transfers energy into or out of a system.

$W = Fd \cos\theta$	Work by constant force F at angle θ to displacement d
$W = \int \mathbf{F} \cdot d\mathbf{x}$	Work by variable force — area under F-x graph
$KE = \frac{1}{2}mv^2$	Kinetic energy of a particle
$W_{\text{net}} = \Delta KE$	Work-Kinetic Energy Theorem
$U_g = mgy$	Gravitational potential energy (y measured upward)
$U_s = \frac{1}{2}kx^2$	Spring (elastic) potential energy; k = spring constant

- **Conservative force:** work done is path-independent (gravity, spring). You can define a potential energy for it.

- **Non-conservative force:** work depends on path (friction, air drag). Energy is converted to thermal energy.
- **Power:** $P = dW/dt = \mathbf{F} \cdot \mathbf{v}$ (watts, $W = J/s$)
 - ★ *Work can be negative (force opposes motion). Negative work removes kinetic energy.*

Chapter 8 — Conservation of Energy

Energy is conserved — it can transform between types but the total within an isolated system is constant.

Energy Conservation Equations

Isolated system (no non-conservative forces): $KE_i + PE_i = KE_f + PE_f$

Non-isolated system: $\Delta E_{\text{mech}} = W_{\text{ext}}$

With friction: $\Delta E_{\text{mech}} = -f_k d$ (energy lost to heat)

General: $\Delta E_{\text{system}} = \Sigma T$ (T = all energy transfers across boundary)

- **Isolated system:** no energy crosses the boundary $\rightarrow$ total mechanical energy $E_{\text{mech}} = KE + PE = \text{constant}$.
- **Strategy:** (1) Define system. (2) Identify initial and final states. (3) List all energy terms. (4) Write energy equation. (5) Solve.
- **Power:** $P = \Delta E/\Delta t$ — average rate of energy transfer. $1 \text{ hp} = 746 \text{ W}$.

★ *When friction acts, mechanical energy decreases by $f_k \times d$. That energy becomes internal (thermal) energy — it's not lost, just converted.*

UNIT 3 — ELECTROSTATICS

Chapters 23 · 24 · 25

Chapter 23 — Electric Fields

Electric charge is quantised (multiples of $e = 1.602 \times 10^{-19} \text{ C}$) and conserved. Like charges repel; unlike charges attract.

Coulomb's Law	$F = k_e q_1 q_2 / r^2$ (along line joining charges)
k_e	$8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$ (Coulomb constant)
ϵ_0	$8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2)$ (permittivity of free space) $k_e = 1/(4\pi\epsilon_0)$
Electric field E	$E = F/q_0$ (force per unit positive test charge)
Point charge	$E = k_e q /r^2$ (directed away from + charge)

Superposition Principle

For multiple charges: $E_{\text{total}} = E_1 + E_2 + E_3 + \dots$ (vector sum)

Continuous distribution: $E = k_e \int (dq/r^2) \mathbf{r}$

- **Electric field lines:** start on + charges, end on – charges. Density of lines $\propto$ field strength. Lines never cross.
- **Charged particle in uniform E:** acts like projectile with acceleration $a = qE/m$.
- **Conductors:** free electrons redistribute until $E = 0$ inside a conductor in electrostatic equilibrium.

★ Superposition applies to both forces and fields. Treat each source separately, then add vectorially.

Chapter 24 — Gauss's Law

Gauss's Law provides an elegant shortcut to finding E when the charge distribution has sufficient symmetry (spherical, cylindrical, or planar).

Gauss's Law

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

Electric flux: $\Phi_E = E \cdot A \cdot \cos\theta$ (uniform field through flat surface)

Spherical symmetry	$E = k_e Q/r^2$ (outside a sphere of charge Q; same as point charge)
Cylindrical symmetry	$E = \lambda / (2\pi\epsilon_0 r)$ (distance r from long wire, λ = charge per length)
Planar symmetry	$E = \sigma / (2\epsilon_0)$ (infinite plane, σ = surface charge density)
Inside conductor	$E = 0$ in electrostatic equilibrium
Conductor surface	$E = \sigma / \epsilon_0$ just outside surface (perpendicular)

- **Gaussian surface:** any closed imaginary surface. Choose it so E is constant and parallel (or perpendicular) to the surface everywhere.
- **Net flux depends only on enclosed charge.** External charges contribute zero net flux.

★ Gauss's Law is always true, but only useful for finding E when symmetry allows E to come out of the integral.

Chapter 25 — Electric Potential

Electric potential V is potential energy per unit charge. It is a scalar, which makes many calculations easier than working with vector fields.

Potential difference	$\Delta V = V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{s}$
Uniform field	$\Delta V = -Ed$ (d = displacement parallel to E)
Point charge	$V = k_e q/r$ ($V = 0$ at $r \rightarrow \infty$)
Potential energy	$U = qV$ (PE of charge q at point where potential is V)
E from V	$E_x = -dV/dx$ (E is the negative gradient of V)
Work by E	$W = q(V_A - V_B) = -\Delta U$

- **Equipotential surfaces:** surfaces of constant V. E field lines are always perpendicular to them.
- **Superposition for V:** $V_{\text{total}} = \sum k_e q_i/r_i$ (scalar sum — easier than vector sum for E).
- **Conductor:** entire conductor is at the same potential. V is constant inside.
- **Electron volt (eV):** 1 eV = 1.602×10^{-19} J — energy gained by one electron through 1 V potential difference.

★ Moving in the direction of E, potential decreases. A positive charge naturally moves from high V to low V (like a ball rolling downhill).

UNIT 4 — CAPACITORS & DC CIRCUITS

Chapters 26 · 27 · 28

Chapter 26 — Capacitance & Dielectrics

A capacitor stores energy in an electric field between two conductors separated by an insulating gap or dielectric.

Capacitance	$C = Q/V$ (farads, $F = C/V$) — depends only on geometry
Parallel plate	$C = \epsilon_0 A/d$ (area A , separation d , no dielectric)
With dielectric	$C = \kappa \epsilon_0 A/d$ (κ = dielectric constant ≥ 1)
Energy stored	$U = Q^2/(2C) = \frac{1}{2}CV^2 = \frac{1}{2}QV$
Energy density	$u = \frac{1}{2}\epsilon_0 E^2$ (energy per unit volume in E field)

Capacitor Combinations

Series: $1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 + \dots$ (same Q , different V)

Parallel: $C_{eq} = C_1 + C_2 + C_3 + \dots$ (same V , different Q)

- **Series capacitors:** smallest capacitor dominates. Equivalent C is always LESS than the smallest individual C .
- **Parallel capacitors:** equivalent C is always MORE than the largest individual C .
- **Dielectric:** fills space between plates, increases C by factor κ , and reduces E by factor κ . Also prevents breakdown at high voltages.

★ When a dielectric is inserted into an isolated (disconnected) capacitor, Q stays constant but V decreases and U decreases. When connected to a battery, V stays constant, Q and U increase.

Chapter 27 — Current & Resistance

Electric current is the flow of charge. Resistance is opposition to that flow.

Current	$I = dQ/dt$ (amperes, $A = C/s$)
Current density	$J = I/A = nqv_d$ (n = carrier density, v_d = drift speed)
Ohm's Law	$V = IR$ (R in ohms, $\Omega = V/A$)
Resistivity	$R = \rho L/A$ (ρ = resistivity, L = length, A = cross-section area)
Temperature dep.	$\rho = \rho_0[1 + \alpha(T - T_0)]$ (α = temperature coefficient)
Power	$P = IV = I^2R = V^2/R$ (watts)

- **Drift velocity v_d :** very slow ($\sim 10^{-4}$ m/s) — electrons zig-zag through collisions. Signal propagates at near light speed.
 - **Ohmic vs non-ohmic:** Ohm's law holds for many metals at constant T . Diodes, transistors are non-ohmic.
 - **Superconductors:** below critical temperature T_c , resistance drops to exactly zero.
- ★ Good conductors have low ρ (e.g. copper $\rho \approx 1.7 \times 10^{-8} \Omega \cdot m$). Good insulators have high ρ (glass $\rho \sim 10^{11} - 10^{14} \Omega \cdot m$).

Chapter 28 — Direct-Current Circuits

DC circuit analysis uses EMF sources (batteries) and the systematic application of Kirchhoff's rules.

EMF ($\mathcal{E}$)	Energy per unit charge provided by battery; ideal battery: $V = \mathcal{E}$. Real battery: $V = \mathcal{E} - Ir$ (r = internal resistance)
Resistors series	$R_{eq} = R_1 + R_2 + R_3 + \dots$ (same I through each)
Resistors parallel	$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3 + \dots$ (same V across each)

Kirchhoff's Rules

Junction rule (KCL): $\sum I_{in} = \sum I_{out}$ (charge conservation)

Loop rule (KVL): $\sum \Delta V = 0$ around any closed loop (energy conservation)

Sign convention: cross resistor in direction of $I \rightarrow \Delta V = -IR$

Cross battery from $-$ to $+$ $\rightarrow \Delta V = +\mathcal{E}$; from $+$ to $- \rightarrow \Delta V = -\mathcal{E}$

RC Circuits (Charging / Discharging)

Charging: $q(t) = C\mathcal{E}(1 - e^{-t/RC})$, $I(t) = (\mathcal{E}/R)e^{-t/RC}$

Discharging: $q(t) = Q_0 e^{-t/RC}$, $I(t) = I_0 e^{-t/RC}$

Time constant: $\tau = RC$ (time to reach 63% of final charge)

- **Strategy for Kirchhoff:** (1) Assign current directions (guess is fine). (2) Write KCL at junctions. (3) Write KVL for enough independent loops. (4) Solve the system of equations.
- **At $t = 0$ (capacitor uncharged):** behaves like a wire (short circuit). At $t = \infty$: behaves like an open circuit, $I = 0$.

★ Series R gives larger R_{eq} ; parallel R gives smaller R_{eq} . The opposite is true for capacitors.

FORMULA QUICK-REFERENCE SHEET

Quantity	Formula	Units
Kinematic (const. a)	$v = v_i + at \mid x = x_i + v_i t + \frac{1}{2}at^2 \mid v^2 = v_i^2 + 2a\Delta x$	m/s, m
Projectile (horiz)	$x = v_i \cos\theta \cdot t, y = v_i \sin\theta \cdot t - \frac{1}{2}gt^2$	m
Centripetal acc.	$a_c = v^2/r = \omega^2 r$	m/s ²
Newton 2nd law	$\Sigma F = ma$	N = kg·m/s ²
Friction	$f_s \leq \mu_s N, f_k = \mu_k N$	N
Work	$W = Fd \cos\theta \mid W_{net} = \Delta KE$	J
KE	$KE = \frac{1}{2}mv^2$	J
PE (gravity/spring)	$U_g = mgy, U_s = \frac{1}{2}kx^2$	J
Power	$P = W/t = Fv$	W = J/s
Coulomb's Law	$F = k_e q_1 q_2 /r^2, k_e = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$	N
Electric field	$E = F/q = k_e q /r^2$	N/C = V/m
Gauss's Law	$\Phi E = Q_{enc}/\epsilon_0$	N·m ² /C
Electric potential	$V = k_e q/r, \Delta V = -Ed, E = -dV/dx$	V = J/C
Capacitance	$C = Q/V = \epsilon_0 A/d, U = \frac{1}{2}CV^2$	F = C/V, J
Cap. series/parallel	$1/C_{eq} = \Sigma 1/C_i \text{ (series)}, C_{eq} = \Sigma C_i \text{ (parallel)}$	F
Current/Resistance	$I = dQ/dt, V = IR, R = \rho L/A$	A, Ω
Power (electrical)	$P = IV = I^2 R = V^2/R$	W
Res. series/parallel	$R_{eq} = \Sigma R_i \text{ (series)}, 1/R_{eq} = \Sigma 1/R_i \text{ (parallel)}$	Ω
Kirchhoff's rules	$\Sigma I_{in} = \Sigma I_{out} \text{ (junction)}, \Sigma \Delta V = 0 \text{ (loop)}$	—
RC time constant	$\tau = RC, q(t) = C\epsilon(1 - e^{-(t/\tau)})$	s, C

Key Constants

$g = 9.80 \text{ m/s}^2$	$k_e = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$
$e = 1.602 \times 10^{-19} \text{ C}$	$m_e = 9.11 \times 10^{-31} \text{ kg}$	$m_p = 1.67 \times 10^{-27} \text{ kg}$

Good luck on your exam! Remember: draw FBDs for mechanics, draw Gaussian surfaces for electrostatics, and always apply Kirchhoff's rules systematically for circuits.